Project SRS

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Problem Understanding &Analysis	7					
Implementation	8					
Task Efferently	10					
					Total obtained mark	
Comments						

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1. Introduction

This document describes what the Statute-Grounded Bangla Legal Assistant must do and how well it must do it. The system is a Bangla-first web application for ordinary citizens of Bangladesh. A user can type a legal question in everyday Bangla and receive a short, plain-Bangla answer that cites the exact Act and section and shows the original statute text beside the explanation. The user can also upload or paste a document or a case description. The system then extracts the legal issue and key facts, compares claims and clauses with the statute text, flags possible contradictions and missing information, links facts to law and to the user's evidence, prepares a template-based Bangla draft, lists the next steps stated in the Acts, and produces a structured case summary. Every output carries an evidence-status indicator and a legal-information disclaimer. The system provides legal information only. It does not give legal advice.

1.1 Problem Statement

When an ordinary person faces a legal problem, for example a worker whose wages are unpaid or a consumer sold an expired product, the usual routes are to ask a lawyer, ask friends or relatives, search Facebook groups or general web pages, or read the Act directly. A lawyer's time costs money, and Acts are long and written in formal legal Bangla. One situation may involve several sections spread across different Acts, and the official Bangladesh Laws portal supports only plain keyword search, which fails when the problem is described in everyday words. People also hold documents (appointment letters, notices, termination letters, receipts, complaint drafts) that they cannot easily check against the law, and they often do not know what to write in a complaint or where to go next.

General chatbots such as ChatGPT and Gemini can answer in Bangla, but they are not tied to the actual text of Bangladeshi statutes. They may quote a section that does not exist, mix Bangladeshi law with Indian or Pakistani law, or use an outdated version, and they cannot show the exact legal text behind an answer. Existing local legal websites are mostly static article collections, and most tools are English-first. No freely available tool combines question answering, document checking, drafting help and next-step guidance for Bangladeshi statutes in Bangla.

AI is needed because the manual steps involved can be done faster and more consistently by software: matching a question written in everyday Bangla to the right sections even when no words are shared (semantic search), rewriting formal legal text as a short plain-Bangla explanation (a language model), extracting the legal issue, facts, claims, dates and amounts from a question or document (LLM-based extraction), and comparing those items with retrieved sections to flag consistency, contradiction and missing information. Keyword or rule-based methods cannot do this reliably because the same situation can be described in many different ways. Without these, the system would only be a list of Acts, equivalent to the current portal.

1.2 Purpose

The primary objectives of the software are:

- To answer legal questions in plain Bangla using only retrieved sections of selected Bangladeshi Acts, with the Act name, section number and original text shown for every answer.

- To verify every citation against the retrieved set and show a supported / partly supported / not supported status with every output.
- To refuse to answer, and say so clearly, when the knowledge base does not contain sufficient support.
- To extract the legal issue, key facts, parties, dates and amounts from a question or an uploaded document.
- To compare the claims and clauses of a document with the statute text and flag possible contradictions and missing information or evidence.
- To help the user prepare a basic Bangla draft (complaint, application or notice) from a template and to list the next steps that the selected Acts actually state.
- To map the facts of a case to statute sections and to the user's evidence, build an evidence matrix and generate a structured Bangla case summary.
- To evaluate all AI features honestly on a self-prepared Bangla test set.

1.3 Scope

The system will:

- Run as a Bangla-first web application, usable on desktop and mobile browsers, with a script-based administration process.
- Use a knowledge base built from the official text of a limited set of Acts (initially four to five; proposed: Bangladesh Labour Act 2006, Consumer Rights Protection Act 2009, Cyber Security Act 2023, Domestic Violence (Prevention and Protection) Act 2010, and selected chapters of the Penal Code 1860). The final list will be fixed in the first two weeks according to text availability and workload.
- Accept questions typed in Bangla (Banglish and English questions on a best-effort basis).
- Provide the core AI features (F1-F4): Bangla legal Q&A, statute-grounded RAG, exact section citation with source verification, and legal issue extraction.
- Provide document and case intelligence (F5-F8): upload or paste of text-based documents (typed PDF, DOCX, TXT, limited length), a verification engine, a contradiction detector and a missing-information / evidence detector.
- Provide legal assistance (F9-F10): template-based Bangla drafting for a small set of common documents and a next-step guide based on procedures found in the selected Acts.
- Provide the trust layer (F11-F12): an evidence-status indicator for every output, and a finder for related provisions and similar example scenarios from a curated library.
- Provide case analysis and reporting (F13-F15): fact-law-evidence mapping, an evidence matrix and a structured Bangla case summary that can be shown or downloaded.
- Show a 'not enough information' response when retrieval confidence is low, a helpful / not helpful feedback button, the statute snapshot date and a mandatory legal-information disclaimer.
- Include an evaluation module with a test set of at least 100 questions and about 30 sample documents.

The system will not:

- Give legal advice, file documents with any authority, or predict the outcome of a case.

- Authenticate documents (detect forgery or fake signatures) or decide whether the facts told by a user are true. 'Verification' in this project means comparison with the statute text only.
- Cover court judgments, case law, rules and regulations, gazette notifications, or Acts outside the selected list. The similar-scenario feature uses curated examples, not real court cases.
- Support OCR of scanned or handwritten documents, voice input, or a mobile application in this version. These are future work.
- Judge the strength, admissibility or genuineness of evidence. The evidence matrix only records what the user has provided.
- Produce documents that are ready to file. All drafts are templates that the user must review.
- Track amendments in real time. The knowledge base is a fixed snapshot with its date shown.
- Provide user accounts or personal case files. Uploaded documents are processed in the session and are not stored permanently.

AI boundaries

The AI will understand a question or document, extract the legal issue and key facts, find relevant sections, summarise them in simple Bangla, cite them, compare claims with the statute text, flag possible contradictions and missing information, fill a draft template, list statute-supported next steps, show an evidence status, and refuse when unsupported. The AI will not decide who is legally right or wrong, judge the truth of a document or of the user's facts, recommend legal action such as 'file a case against X', or answer from knowledge outside the retrieved statute text. Anything that needs a lawyer's judgement is outside the system boundary, and the user is directed to a qualified lawyer or legal-aid service.

2. Design and Implementation Constraints

2.1 Programming Language

Part	Technology	Why we chose it
Backend and AI pipeline	Python with FastAPI	One language for services, data processing and AI code; FastAPI is light, fast and gives automatic API documentation.
Frontend	Streamlit for the prototype, or React with HTML/CSS/JavaScript for a custom interface (final choice by team skill)	Streamlit gives a working Bangla interface quickly; React allows a custom layout if time permits.
Data processing	Python with Requests, BeautifulSoup, regular expressions and Pandas	Collection, cleaning and section-wise splitting of the statute text.
Document handling	pypdf or pdfplumber, python-docx	Reading text-based PDF and DOCX files and exporting drafts and matrices.

Part	Technology	Why we chose it
Data files	JSON and CSV	Templates, issue checklists and the scenario library are easy to edit and review.

2.2 Database & Servers

- A vector database (ChromaDB or FAISS) stores the embeddings of all statute sections together with metadata (Act, chapter, section number, title).
- SQLite stores feedback and logs. Questions and feedback are stored without personal identifiers, for evaluation only.
- Draft templates, issue checklists and the scenario library are kept as JSON/CSV files under Git version control.
- Uploaded documents are processed in memory during the session and deleted when the session ends.
- The browser talks to the FastAPI backend over HTTPS. API keys are kept in environment variables and never in the source code.
- The demo is hosted on a local machine or a free-tier cloud service (for example Hugging Face Spaces or Render). The official Bangladesh Laws portal (bdlaws.minlaw.gov.bd) is the source of the statute text and is used with attribution and in line with its terms of use.

2.3 AI Model & Services

AI feature	Approach and service	Justification
Semantic retrieval (F2, F12)	Open multilingual embedding model with Bangla support (for example multilingual-e5 or BGE-M3 via Sentence-Transformers), vector search for the top-k sections, optional multilingual cross-encoder re-ranking.	A user's everyday wording rarely matches legal wording, so keyword search fails. The best model will be chosen by testing on our own question set.
Grounded answer generation (F1, F3)	Bangla-capable LLM through an API, prompted to use only the retrieved sections, followed by a citation check against the retrieved set.	Grounding keeps answers tied to the official text and reduces invented sections.
Issue, fact and claim extraction (F4, F5)	LLM with constrained prompts and structured (JSON) output.	Produces issue category, facts, parties, dates and amounts in a form the rest of the pipeline can use.
Claim comparison and contradiction detection (F6, F7)	Prompt-based comparison of extracted claims with retrieved	Rules catch exact date and amount conflicts; the LLM

AI feature	Approach and service	Justification
	sections, supported by simple rule checks for dates and amounts.	handles the meaning of clauses.
Missing-information detection (F8)	Comparison of extracted items with a stored checklist for the identified issue.	Checklists written and reviewed by the team keep the result predictable.
Drafting and next steps (F9, F10)	Template-based generation filled by the LLM from the user's details and retrieved text.	Templates limit invention; next steps are listed only if the retrieved sections state them.
Mapping, matrix and summary (F13-F15)	LLM plus retrieval over the extracted facts, the retrieved sections and the user's evidence list.	Reuses the same evidence as all other features, so one pipeline serves all fifteen features.
Evidence status (F11)	Rule-based score from retrieval scores, verification result and the number of supporting sections.	A transparent rule is easy to explain and test.

3. User Classes and Characteristics

User class	Characteristics	Permissions
General Citizen	Main user: worker, consumer, woman facing a domestic problem, or internet user facing online harassment. Often has no legal training and prefers simple Bangla.	Ask questions, upload or paste a document or case description, confirm extracted facts, list evidence, view answers, evidence status and original text, create and download drafts, matrices and summaries, give feedback. No registration is required.
Law Student / Teacher	Needs to locate provisions quickly and see the exact text and related provisions.	Same as a citizen; uses the Act filter and the related-provision finder.
Legal-aid Volunteer / NGO Worker	Answers basic legal queries repeatedly; moderate technical skill.	Same as a citizen; uses checklists, drafts and case summaries to prepare first-level information.

User class	Characteristics	Permissions
Lawyer Reviewer (optional)	Reviews sample answers, drafts and checklists to make sure information is not misleading.	Read-only access to sample outputs and evaluation results supplied by the team.
System Administrator (project team)	Technical team member who maintains the knowledge base and hosting.	Add or update statute data, templates, checklists and the scenario library; rebuild the index by script; view feedback and logs; run the evaluation script. Does not edit a user's output.

4. Requirement Elicitation

This section explains how the requirements in this document were collected from real users and a domain expert, and how they changed as a result. Requirements were elicited through an online questionnaire (Google Forms) for prospective users and an in-person consultation with a law faculty member.

4.1 Elicitation Techniques

Technique	Who took part	Purpose	When
Online questionnaire (Google Forms)	9 respondents (responses received up to 11/10/2026), all university or college students	Measure how people look for legal information, how difficult it is, and how useful each group of features (F1–F15) would be; 20 questions in five sections (Appendix A); response summary in Appendix D	Form open 09/10/2026 – 12/10/2026; responses received on 11/10/2026
Expert consultation with law faculty	Samir Bin Shibbir, Lecturer, Daffodil International University (law faculty)	Learn what makes a legal information tool good or harmful, and check whether the proposed features match what legal practice needs (Appendix B)	10/10/2026, in person
Prototype review / user testing	Classmates and, if possible, the faculty member	Check that answers, evidence status and disclaimers are understood (planned)	Planned – week 13

4.2 Elicitation Evidence

Proof that elicitation took place is attached as appendices: the questionnaire (Appendix A), the expert consultation record (Appendix B), the evidence checklist (Appendix C) and the response summary of the questionnaire (Appendix D).

No.	Evidence item	Reference
E1	Google Forms questionnaire (20 questions)	Appendix A
E2	Questionnaire response summary (9 responses, counts per question)	Appendix D
E3	Record of the in-person consultation with the law faculty member	Appendix B

4.3 Key Findings

A. Expert consultation (law faculty)

The proposed feature set was presented to Samir Bin Shibbir, Lecturer. His feedback is summarised below (paraphrased from the consultation).

Theme	What the faculty member said	Effect on requirements
Exact provisions	The result must match the exact section (dhara) of the Act, not a general explanation.	Confirms FR-4, FR-5 and AIR-2; FR-4 revised to also show the source (Section 4.4).
Updates and sources	Users should be able to find updates to the law, and every piece of information should come with its source.	FR-4 and FR-30 revised: source shown with every section; snapshot date and a pointer to the official portal for later amendments (Section 4.4).
What makes the tool good	Showing the exact section, refusing to answer when the law does not support the question, and helping to draft complaints.	Confirms FR-5, FR-6, FR-17 and FR-18.
What makes the tool harmful	Giving legal advice, and invented (hallucinated) content.	Confirms AIR-5 (neutral informational style), AIR-2 (citations must be in the retrieved set) and the out-of-scope limits in Section 1.3.
Feature set	The faculty member liked the proposed features; no feature was asked to be added, removed or changed.	No change to the feature list F1–F15.

B. Questionnaire (Google Forms)

The questionnaire received 9 responses, all from university or college students. Results are summarised below by section and linked to the features the questions were written for; counts per question are in Appendix D.

Section / questions	Features	Result (n = 9)
A (Q1–Q4)	Need and problem	7 of 9 (78%) have needed legal information in Bangladesh. Search engines (5) and AI chatbots (3) are the main sources. 6 of 9 (67%) find it very or somewhat difficult to find the correct section; 8 of 9 report some difficulty.
B (Q5–Q7)	F1, F2, F3, F11	Simple Bangla explanation was rated 4–5 (very / extremely important) by 6 of 9 (67%). Showing the Act, section number and original text was rated 4–5 by 7 of 9 (78%). 5 of 9 are extremely concerned that AI answers may be incorrect, outdated or based on another country's law; all 9 report at least some concern.
C (Q8–Q12)	F4–F9	Document explanation in Bangla: 7 of 9 (78%) extremely or somewhat useful. Consistency check with sources: 8 of 9 (89%) definitely or probably useful. Contradiction detection: 8 of 9 (89%) extremely or somewhat useful. Missing-information checklist: 9 of 9 (100%) helpful. Bangla draft: 9 of 9 (100%) would definitely or probably use it.
D (Q13–Q18)	F10–F15	Next steps supported by law: 7 of 9 (78%) rated 4–5. Related laws, examples and case summary: 8 of 9 (89%) extremely useful. Evidence status indicator: 9 of 9 (100%) useful (7 extremely). Facts–law–evidence structure: 9 of 9 (100%) helpful (8 very helpful). Evidence matrix: 8 of 9 (89%) extremely or somewhat useful. Most important feature (Q15): reliable answers with evidence and source indicators (3), document analysis and checklists (2), simple Bangla explanations (2), exact citations (1), all features (1).
E (Q19–Q20)	F3, F11	Verifying an AI answer against the original Act text was rated 4–5 by 8 of 9 (89%). Biggest concern: incorrect or misleading legal information (6 of 9, 67%), then over-reliance on AI instead of professionals (2) and privacy of uploaded documents (1).

Limitation. The sample is small (9 responses) and consists only of university and college students, so the percentages are indicative rather than representative. The expert view is covered by the law faculty consultation; further responses from legal-aid volunteers and other groups are planned for the evaluation stage.

Summary. Every proposed feature group was rated useful by most respondents, the strongest support being for the evidence status indicator, the facts–law–evidence structure, the missing-information

checklist and Bangla drafting. The main concern – incorrect or misleading answers – directly supports the design choices of grounding in the statute text, citation verification and refusal when unsupported.

4.4 Requirement Changes After Elicitation

Requirements that were added, modified or removed after collecting user input are recorded below with the reason for each change. Requirement IDs refer to Section 5.

ID	Change	Before Elicitation	After Elicitation	Source / Reason
FR-4	Modified	The system shall display, for every answer, the Act name, section number and original section text it is based on.	The system shall display, for every answer, the Act name, section number, original section text and the source of that text (official Bangladesh Laws portal).	Faculty consultation: every piece of information must carry its source and match the exact section. Supported by questionnaire Q6 and Q19.
FR-30	Modified	The system shall display the date of the statute snapshot used.	The system shall display the date of the statute snapshot used and state that later amendments may exist, with a pointer to the official portal for the latest text.	Faculty consultation: users should be able to find updates to the law. Supported by questionnaire Q7 (concern about outdated answers).
–	Added / removed	–	None	Faculty consultation: the proposed features were liked; questionnaire: all feature groups rated useful by most respondents.

Requirements confirmed by elicitation (no change):

Requirement	Why it was confirmed	Source
FR-5, AIR-2 (citation check, no invented sections)	Hallucinated content was named as the main harm to avoid; incorrect information is the top concern (6 of 9).	Faculty consultation; Q7, Q19, Q20
FR-6 (refuse when unsupported)	Refusal was named as good practice.	Faculty consultation
FR-16 (missing-information checklist)	Checklist rated helpful by 9 of 9.	Q11

Requirement	Why it was confirmed	Source
FR-17, FR-18 (Bangla drafting)	Drafting help named as valuable; 9 of 9 would use it.	Faculty consultation; Q12
FR-20 (evidence status)	Indicator rated useful by 9 of 9 (7 extremely).	Q16
FR-23, FR-25 (fact–law–evidence mapping, evidence matrix)	Structure rated helpful by 9 of 9; matrix useful for 8 of 9.	Q17, Q18
AIR-5 (no legal advice)	Giving legal advice was named as harmful.	Faculty consultation

5. Functional Requirements (FR)

Requirements that use AI are marked (AI). Priority follows the three tiers of the project plan: Tier 1 (F1-F4, F11) High; Tier 2 (F5, F6, F8, F10, F13) High or Medium; Tier 3 (F7, F9, F12, F14, F15) Medium or Low.

ID	Requirement Name	Description	Users	Priority
FR-1	Bangla Question Input (F1)	The system must accept a question typed in Bangla through the web interface (Banglish and English on a best-effort basis).	All users	High
FR-2	Semantic Retrieval (AI) (F2)	The system must retrieve the top-k (default k = 5) most relevant statute sections for a question.	All users	High
FR-3	Grounded Answer (AI) (F1)	The system must generate an answer in simple Bangla using only the retrieved sections.	All users	High
FR-4	Exact Citation (F3)	The system must display, for every answer, the Act	All users	High

ID	Requirement Name	Description	Users	Priority
		name, section number and original section text it is based on.		
FR-5	Citation Verification (AI) (F3)	The system must verify that every section cited in an output is among the retrieved sections and must block outputs that fail the check.	All users	High
FR-6	Low-Confidence Refusal (AI) (F2)	The system must display a 'cannot find a supported answer' message when the best retrieval score is below a configured threshold.	All users	High
FR-7	Act Filter (F2)	The system must allow the user to filter the search by Act (optional).	All users	Low
FR-8	Legal Issue Extraction (AI) (F4)	The system must extract, from a question or document, the legal issue category, key facts, parties, dates and amounts, and display them for the user to confirm or correct.	All users	High

ID	Requirement Name	Description	Users	Priority
FR-9	Document Upload (F5)	The system must allow the user to upload a text-based PDF, DOCX or TXT file (up to 10 pages / 5 MB) or paste text.	All users	Medium
FR-10	Document Summary (AI) (F5)	The system must show a Bangla summary of the key clauses, dates, amounts and claims found in the document.	All users	Medium
FR-11	Unreadable File Handling (F5)	The system must tell the user clearly when a file cannot be read (for example a scanned image) instead of producing an analysis.	All users	Medium
FR-12	Claim Verification (AI) (F6)	The system must compare each extracted claim or clause with the retrieved sections and label it as consistent, conflicts with, or not covered, with the cited section.	All users	High
FR-13	Verification Notice (F6)	The system must state that verification means comparison with the statute snapshot and	All users	High

ID	Requirement Name	Description	Users	Priority
		does not confirm that a document is genuine or that the facts are true.		
FR-14	Internal Contradiction Detection (AI) (F7)	The system must flag possible contradictions between dates, amounts or statements inside a document or case description and show the conflicting text side by side.	All users	Medium
FR-15	Statute Conflict Flags (AI) (F7)	The system must flag possible conflicts between a document and the statute text and mark every flag as 'possible' for user review.	All users	Medium
FR-16	Missing Information Detection (AI) (F8)	The system must compare the user's input with a stored checklist for the identified issue and list the items found and the items not found.	All users	Medium
FR-17	Draft Templates (F9)	The system must provide at least three Bangla draft templates (for example unpaid-wage complaint, consumer	All users	Medium

ID	Requirement Name	Description	Users	Priority
		complaint, online-harassment complaint) and fill them using the user's details.		
FR-18	Draft Placeholders and Export (F9)	The system must mark missing details in a draft with clear placeholders, include cited sections, and allow the draft to be edited and downloaded.	All users	Medium
FR-19	Next-Step Guide (AI) (F10)	The system must list next steps only when they are stated in the retrieved sections; otherwise it must refer the user to a qualified lawyer or legal-aid service.	All users	High
FR-20	Evidence Status (AI) (F11)	The system must display an evidence status (supported / partly supported / not supported) with a short reason for every answer and analysis output.	All users	High
FR-21	Related Provisions (F12)	The system must show related sections from	All users	Low

ID	Requirement Name	Description	Users	Priority
		other selected Acts for a retrieved section.		
FR-22	Similar Scenarios (F12)	The system must show up to three similar example scenarios from the curated library, labelled clearly as examples and not real court cases.	All users	Low
FR-23	Fact-Law Mapping (AI) (F13)	The system must produce a table that links each extracted fact to the retrieved section that applies to it, with a citation, or marks the fact as 'no applicable section found'.	All users	High
FR-24	Evidence Listing (F13)	The system must let the user list the evidence they have (for example appointment letter, payslip, receipt, screenshot, message) by selecting from a list or typing, and link each item to the related fact.	All users	Medium
FR-25	Evidence Matrix (F14)	The system must generate a matrix in which each row	All users	Low

ID	Requirement Name	Description	Users	Priority
		is a legal point or fact and each column is an evidence item, with cells marked provided, missing or unclear.		
FR-26	Matrix Download (F14)	The system must allow the matrix to be downloaded as DOCX or CSV and must state that it records only what the user has provided.	All users	Low
FR-27	Case Summary (AI) (F15)	The system must generate a structured Bangla case summary containing parties, key facts, legal issue, relevant sections with citations, evidence status and missing items.	All users	Low
FR-28	Summary Grounding and Export (F15)	The system must build the case summary only from the user's input and the system's verified outputs and must allow it to be edited and downloaded.	All users	Low
FR-29	Disclaimer	The system must show a legal-information	All users	High

ID	Requirement Name	Description	Users	Priority
		disclaimer with every output.		
FR-30	Snapshot Date	The system must display the date of the statute snapshot used.	All users	High
FR-31	Feedback	The system must allow the user to mark an output as helpful or not helpful and optionally add a comment.	All users	Medium
FR-32	Data Handling	The system must store questions and feedback without personal identifiers, for evaluation only, and must delete uploaded documents when the session ends.	All users, Admin	High
FR-33	Knowledge Base Administration	An administrator must be able to add or update the statute data, templates, checklists and scenario library and rebuild the index using a script.	Admin	Medium
FR-34	Evaluation Script	The system must include an evaluation script that reports Hit@k, MRR, faithfulness and	Admin	Medium

ID	Requirement Name	Description	Users	Priority
		the results of the document-feature tests on the test sets.		

6. Non-Functional Requirements

6.1 AI Quality Requirements

The targets below are to be reported honestly after evaluation, including weak areas.

ID	Feature	Requirement
AIR-1	F2	A correct relevant section shall appear in the top 5 results for at least 75% of test questions.
AIR-2	F3	At least 90% of section citations in generated outputs shall refer to sections that exist in the retrieved context.
AIR-3	F6, F11	The system shall refuse or state insufficient information for at least 80% of out-of-scope test questions (unrelated topics or un-indexed Acts).
AIR-4	F1	Answers shall be in Bangla and not exceed about 200 words, excluding the quoted section text.
AIR-5	All	The system shall not give instructions such as 'file a case against X' or predict outcomes; the prompt and an output check shall enforce a neutral, informational style.
AIR-6	F4	The issue category shall be correct for at least 80% of labelled test questions and sample documents.

ID	Feature	Requirement
AIR-7	F5, F6	At least 70% of key dates and amounts shall be extracted correctly, and at least 70% of claim labels shall match the manual labels.
AIR-8	F7	The detector shall find at least 60% of planted contradictions; false alarms shall be reported.
AIR-9	F8	At least 80% of checklist items marked missing shall match the manual judgement on the sample documents.
AIR-10	F9, F10	Every draft and next-step list shall contain only sections in the retrieved set and shall be checked by the team against the source text on at least 20 outputs.
AIR-11	F11	Outputs labelled 'supported' shall have a higher manual-faithfulness rate than those labelled 'partly supported' or 'not supported'.
AIR-12	F12	For at least 70% of test questions, a human judge shall consider at least one of the top 3 related provisions or scenarios relevant.
AIR-13	F13	At least 70% of extracted facts shall be linked to a correct section, and facts with no applicable section shall be marked as such rather than forced to a section.
AIR-14	F14	At least 75% of evidence-matrix cells shall match the manual labels on the sample cases.

ID	Feature	Requirement
AIR-15	F15	Case summaries shall include the key parties, dates, amounts and issue in at least 80% of sample cases, and a manual check of at least 20 summaries shall find no statement unsupported by the input or the cited sections.

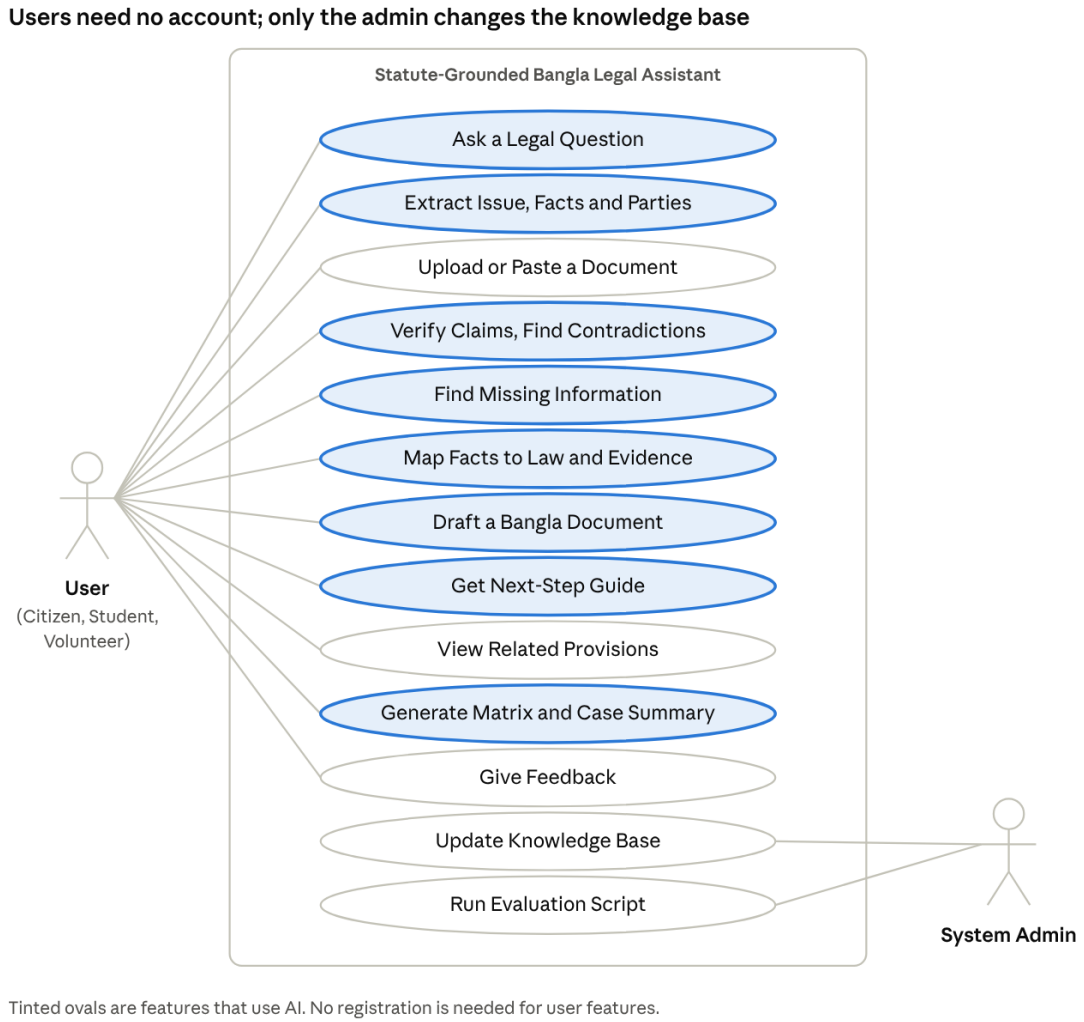
6.2 Other Non-Functional Requirements

- **Performance:** On a normal internet connection, a typical answer must appear within 10 seconds and a document analysis of up to 10 pages within 60 seconds.
- **Reliability and Fallback:** If the LLM API, the retriever or the file reader fails, the system must show a clear error message and must never show an unverified answer. If retrieval confidence is low, it must refuse rather than guess.
- **Security:** API keys must be kept in environment variables, all communication must use HTTPS, and uploaded file types and sizes must be validated.
- **Privacy:** No names, phone numbers or account data are collected. Users are warned not to enter or upload personal details. Uploaded files are processed in the session and not kept. Questions are stored without identifiers.
- **Usability:** The interface must be simple, display Bangla fonts correctly and be usable on mobile browsers.
- **Transparency:** The sources, the evidence status and the statute snapshot date must always be visible, and flags for contradictions and missing information must be shown as 'possible'.
- **Localization:** All screens, answers, drafts and summaries must be available in Bangla.
- **Compatibility:** The web application must run on current versions of common desktop and mobile browsers.
- **Data Integrity:** Every section in the knowledge base must keep its Act, chapter and section metadata, and the snapshot date must be recorded with each rebuild of the index.
- **Maintainability:** Data preparation, indexing, templates, checklists and evaluation must be separate scripts or files so that statute data can be refreshed. The code must be modular, kept under Git version control and documented.

7. System Design Diagrams

7.1 Use Case Diagram

The diagram shows the two actor groups and the main use cases. Tinted ovals are the features that depend on AI. The user group (citizen, law student, legal-aid volunteer) reaches all question, document, drafting and reporting features without registration. The system administrator maintains the knowledge base and runs the evaluation script.



use cases · 11 user features, 2 admin tasks

Figure 1: Use case diagram of the Statute-Grounded Bangla Legal Assistant

7.2 Use Case Descriptions

Main success flows are given below for the core and High priority features.

UC-01 Ask a Legal Question

Field	Details
Actor	User (retrieval index, LLM and citation verifier as supporting systems)
Related FR	FR-1 to FR-6, FR-20, FR-29, FR-30
Precondition	The statute index has been built and the internet is available.

Field	Details
Main Success Flow	(1) The user types a question in Bangla. (2) The system embeds the question and retrieves the top-k sections, optionally filtered by Act. (3) The best retrieval score is above the threshold, so the LLM writes a short Bangla answer using only the retrieved sections. (4) The system checks that every cited section is in the retrieved set. (5) The system computes the evidence status. (6) The system shows the answer with the Act, section number and original text, the evidence status, the snapshot date and the disclaimer. (7) The user may mark the answer helpful or not helpful.
Alternate / Exception	Best score below the threshold at step 3: the system shows 'cannot find a supported answer' and suggests a qualified lawyer or legal-aid service. Citation check fails at step 4: the output is blocked and an error is shown. LLM API unavailable: the system shows a clear error and no answer.
Postcondition	The user has an answer tied to cited sections with a visible evidence status. No personal data is stored.

UC-02 Extract Legal Issue and Facts

Field	Details
Actor	User (LLM as supporting system)
Related FR	FR-8
Precondition	The user has entered a question, a case description or document text.
Main Success Flow	(1) The system sends the input text to the LLM for structured extraction. (2) The LLM returns the issue category, key facts, parties, dates, amounts and likely Act. (3) The system shows the extracted items. (4) The user confirms or corrects them. (5) The confirmed items guide retrieval for the other features.
Alternate / Exception	Issue cannot be identified: the system asks the user to add details or choose an Act.

Field	Details
Postcondition	The confirmed issue and facts are available for the rest of the session.

UC-03 Analyse a Document and Verify Claims

Field	Details
Actor	User (retrieval index and LLM as supporting systems)
Related FR	FR-9 to FR-15, FR-20
Precondition	The user has a text-based PDF, DOCX or TXT file (up to 10 pages / 5 MB) or pasted text.
Main Success Flow	(1) The user uploads or pastes the document. (2) The system reads the text and shows a Bangla summary of the key clauses, dates, amounts and claims. (3) The issue and facts are extracted (UC-02). (4) The system retrieves the related sections. (5) Each claim or clause is labelled consistent, conflicts with, or not covered, with the cited section. (6) The system flags possible contradictions inside the document and with the statute, shown side by side and marked 'possible'. (7) The system shows the evidence status and the notice that verification means comparison with the statute snapshot only.
Alternate / Exception	File cannot be read (for example a scanned image): the system says so and does not analyse. File too large or wrong type: it is rejected with a message. No relevant section: the claim is marked not covered.
Postcondition	The analysis is shown and the document is deleted when the session ends.

UC-04 Find Missing Information and Map Evidence

Field	Details
Actor	User
Related FR	FR-16, FR-23, FR-24
Precondition	The legal issue has been identified (UC-02).

Field	Details
Main Success Flow	(1) The system compares the input with the stored checklist for the issue. (2) It lists the items found and the items not found. (3) The user lists the evidence they have by selecting from a list or typing. (4) The system links each fact to the applicable section and to the related evidence item, or marks the fact 'no applicable section found'.
Alternate / Exception	No checklist exists for the issue: the system says so and skips the comparison.
Postcondition	A fact-law-evidence table is available for the matrix and the case summary.

UC-05 Draft a Bangla Document

Field	Details
Actor	User (LLM as supporting system)
Related FR	FR-17, FR-18
Precondition	The legal issue has been identified and a template exists for it.
Main Success Flow	(1) The user chooses a template (for example unpaid-wage complaint, consumer complaint or online-harassment complaint). (2) The system fills it from the user's details and the retrieved sections. (3) Missing details are shown as placeholders. (4) The user edits the draft. (5) The user downloads it.
Alternate / Exception	No template for the issue: the system offers the closest template or says none is available.
Postcondition	An editable draft with cited sections exists. Nothing is filed with any authority.

UC-06 Get Next-Step Guide

Field	Details
Actor	User
Related FR	FR-19, FR-20

Field	Details
Precondition	The legal issue has been identified.
Main Success Flow	(1) The system reads the retrieved sections for the issue. (2) Where the sections state a procedure, the system lists the steps in order with the Act and section. (3) The system shows the evidence status and the disclaimer.
Alternate / Exception	No procedure is stated in the retrieved sections: the system refers the user to a qualified lawyer or legal-aid service.
Postcondition	The user knows the steps stated in the Acts or where to seek help.

UC-07 Generate Evidence Matrix and Case Summary

Field	Details
Actor	User (LLM as supporting system)
Related FR	FR-25 to FR-28
Precondition	The fact-law-evidence mapping exists (UC-04).
Main Success Flow	(1) The system builds a matrix of legal points against evidence items with provided, missing or unclear cells. (2) The system generates a one-page Bangla case summary with parties, key facts, legal issue, relevant sections, evidence status and missing items. (3) The user edits and downloads the matrix (DOCX or CSV) and the summary.
Alternate / Exception	The evidence list is empty: all cells show missing.
Postcondition	The matrix and summary are available. The matrix records only what the user has provided.

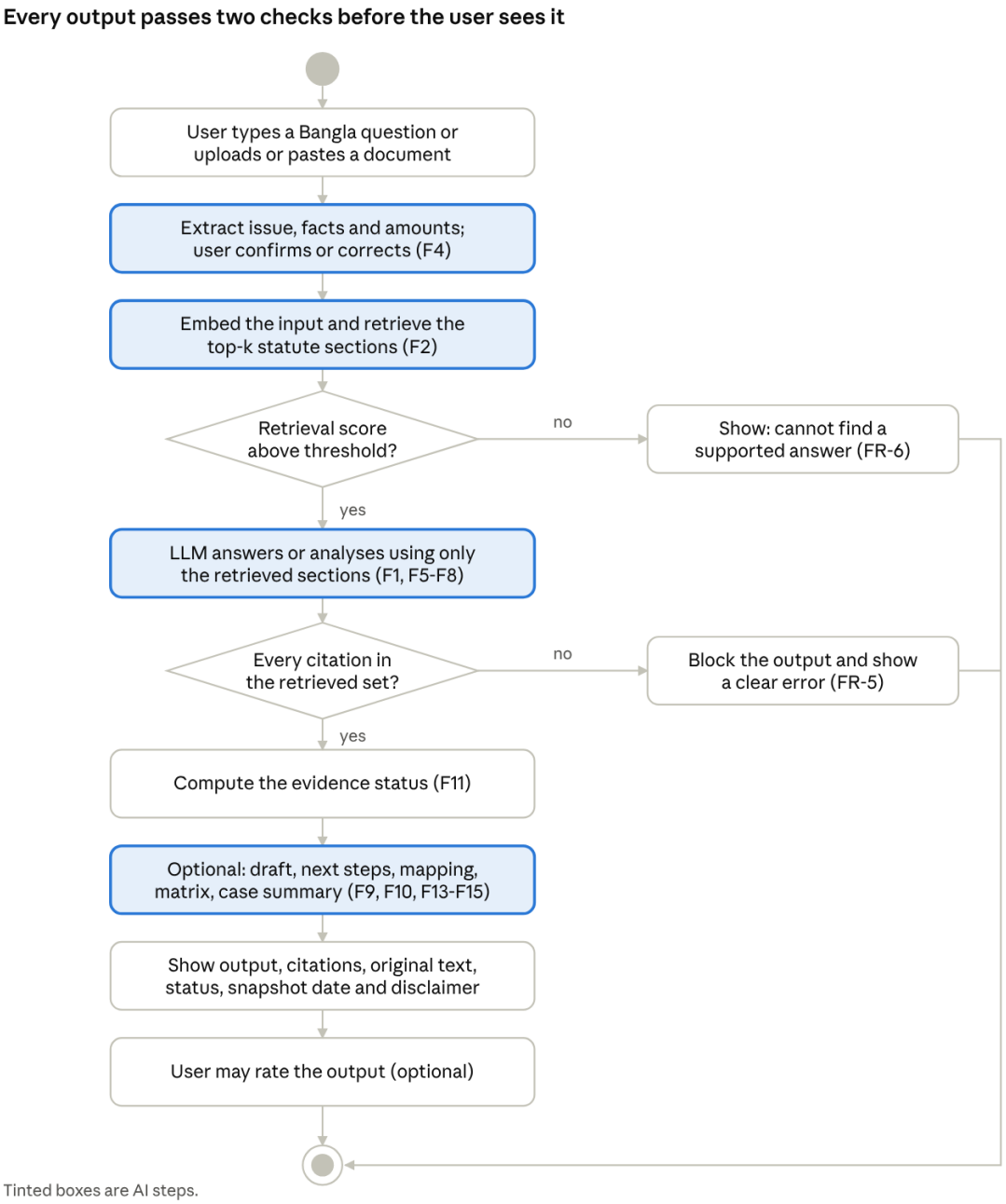
UC-08 Update Knowledge Base

Field	Details
Actor	System Admin
Related FR	FR-33, FR-34
Precondition	The admin has the updated statute text, templates, checklists or scenario library.

Field	Details
Main Success Flow	(1) The admin adds or edits the data files. (2) The admin runs the rebuild script. (3) The system cleans and splits the text section by section, creates embeddings, rebuilds the index and records the snapshot date. (4) The admin runs the evaluation script and checks Hit@k, MRR and faithfulness.
Alternate / Exception	A data file is malformed: the script reports the file and stops.
Postcondition	The index and the snapshot date are updated.

7.3 Activity Diagram

The activity diagram shows the flow of a typical request: from the user's input, through issue extraction, retrieval and grounded generation, to the two safety checks and the displayed output. Optional features such as drafting and the case summary run on the same verified evidence. Tinted boxes are AI steps.

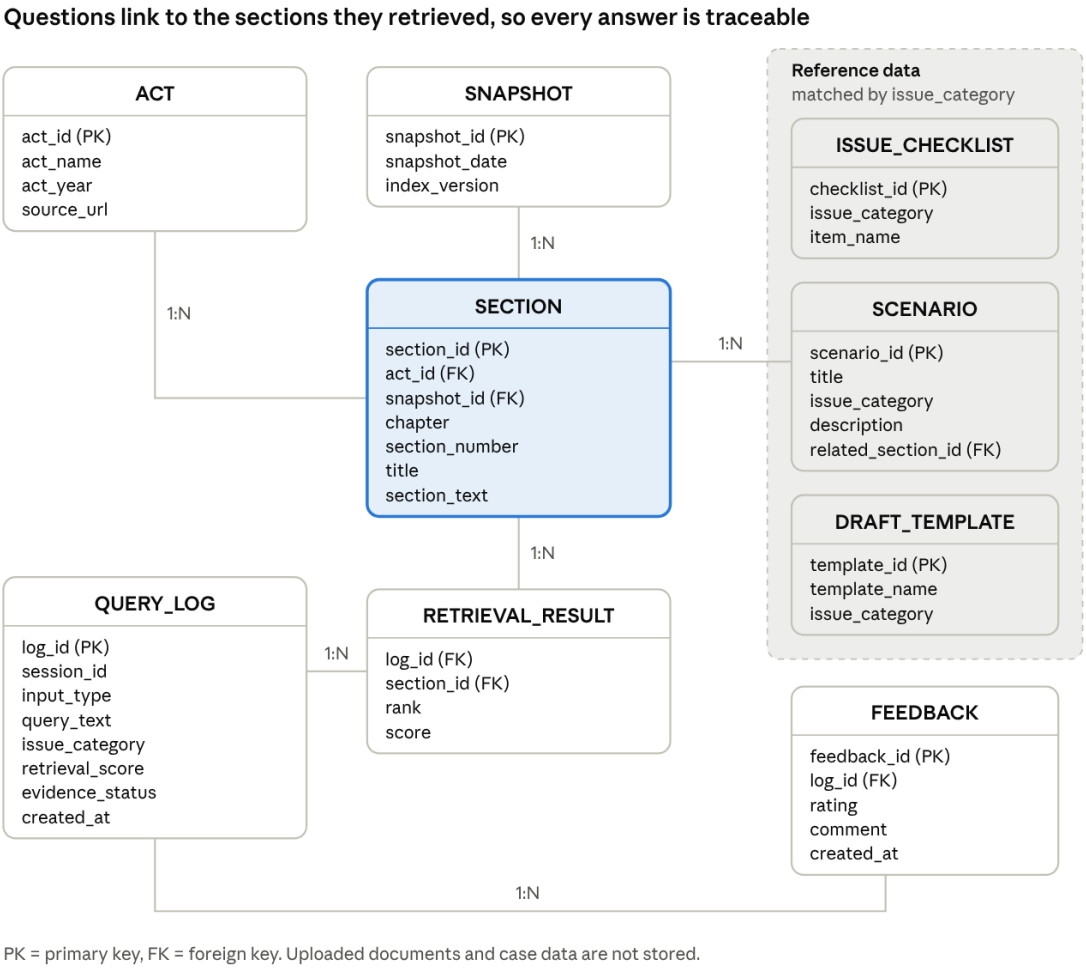


activity flow · 2 decisions, 2 stop paths

Figure 2: Activity diagram of the question and document analysis flow

7.4 Entity Relationship Diagram (ERD)

The database stores the statute knowledge base, the reference data and the evaluation logs. Uploaded documents, extracted facts, evidence lists, matrices and case summaries are session data held in memory and are not stored (FR-32).



ERD · 9 entities, 6 relationships

Figure 3: Entity relationship diagram of the Statute-Grounded Bangla Legal Assistant

Entities and attributes

Entity	Attributes
SNAPSHOT	snapshot_id (PK), snapshot_date, index_version
ACT	act_id (PK), act_name, act_year, source_url
SECTION	section_id (PK), act_id (FK), snapshot_id (FK), chapter, section_number, title, section_text
SCENARIO	scenario_id (PK), title, issue_category, description, related_section_id (FK)
ISSUE_CHECKLIST	checklist_id (PK), issue_category, item_name

Entity	Attributes
DRAFT_TEMPLATE	template_id (PK), template_name, issue_category
QUERY_LOG	log_id (PK), session_id, input_type, query_text, issue_category, retrieval_score, evidence_status, created_at
RETRIEVAL_RESULT	log_id (FK), section_id (FK), rank, score
FEEDBACK	feedback_id (PK), log_id (FK), rating, comment, created_at

Relationships and cardinality

- One ACT contains many SECTION records (1:N).
- One SNAPSHOT covers many SECTION records (1:N), so every section belongs to a dated snapshot.
- One SECTION can be linked to many SCENARIO records (1:N).
- QUERY_LOG and SECTION have a many-to-many relationship (M:N), resolved through RETRIEVAL_RESULT: one query has many retrieval results (1:N), and one section appears in many retrieval results (1:N).
- One QUERY_LOG record can receive many FEEDBACK records (1:N).
- ISSUE_CHECKLIST and DRAFT_TEMPLATE have no foreign key. They are matched to a question or document through issue_category.
- session_id is a random value for one browser session. It is not linked to a person, and no names, phone numbers or account data are stored.

7.5 AI Feature Design

AI is the core of this system. This section describes the data, the models, the pipeline, the integration and the evaluation of the fifteen AI features, which share one retrieval pipeline.

Data

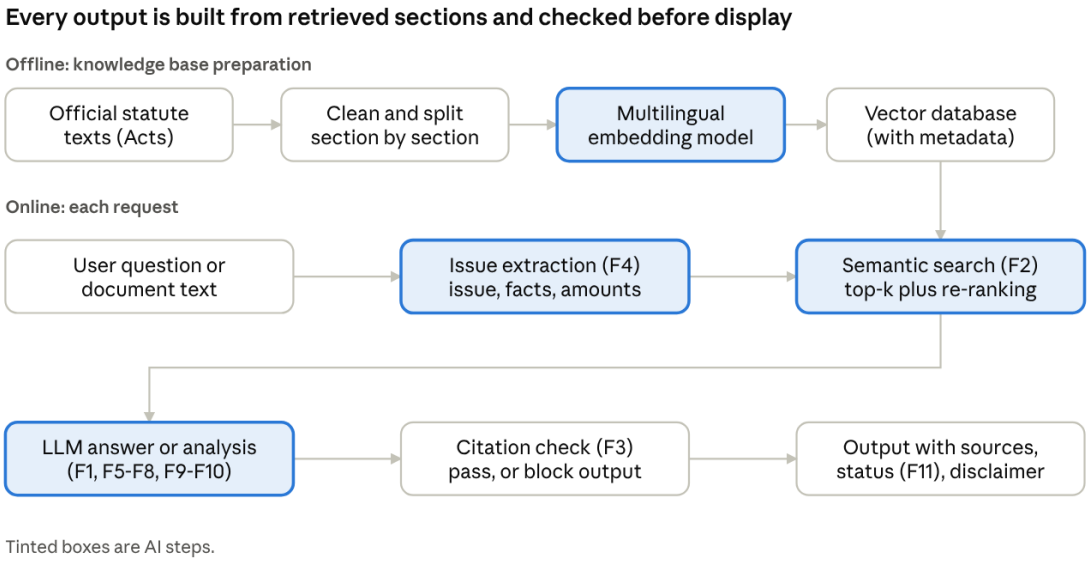
- **Statute corpus:** the official text of the selected Acts from the Bangladesh Laws portal, cleaned and split section by section, with Act, chapter, section number and title kept as metadata. [number of Acts and sections when known]
- **Question set:** at least 100 Bangla questions in everyday wording, including out-of-scope questions, each labelled with the relevant sections and the issue category. [number when known]
- **Sample documents:** about 30 text-based documents (appointment letters, notices, termination letters, receipts, complaint drafts), labelled with key dates and amounts, claim labels, planted contradictions and missing items. [number when known]
- **Templates, checklists and scenario library:** prepared by the team, cross-checked against the source text and, where possible, reviewed by a lawyer.

- **Preparation:** headers, footers and formatting noise are removed, Bangla Unicode text is kept, and each section becomes one retrievable unit with its metadata.

Model

- **Retrieval (F2, F12):** a multilingual embedding model with Bangla support (multilingual-e5 or BGE-M3), chosen by testing on our question set, with optional cross-encoder re-ranking. The baseline is plain keyword search over the same sections.
- **Generation (F1, F3):** a Bangla-capable LLM through an API, prompted to use only the retrieved sections. The baseline is the same LLM without retrieval, to measure the gain in faithfulness.
- **Extraction and comparison (F4-F8, F13):** an LLM with structured (JSON) output, supported by simple rule checks for dates and amounts. The reference is the manual labels on the sample documents.
- **Evidence status (F11):** a rule-based score from the retrieval scores, the verification result and the number of supporting sections.

Pipeline



RAG pipeline · 4 offline steps, 6 online steps

Figure 4: RAG pipeline and offline knowledge base preparation

Offline, the statute text is cleaned, split section by section, embedded and stored in the vector database. Online, the input is analysed for its legal issue and facts, the most similar sections are retrieved, the LLM writes an answer or analysis from those sections only, the citation check confirms that every cited section was retrieved, and the system adds the evidence status and the disclaimer. The same retrieved evidence is reused by drafting, next steps, mapping, the evidence matrix and the case summary.

Integration

The web interface sends the question or document to the FastAPI backend. The backend reads the file, runs the issue extraction, queries the vector database, calls the LLM with only the retrieved sections, runs the citation check, computes the evidence status and returns the output with the sources, the status, the snapshot date and the disclaimer. Feedback is saved to SQLite without personal identifiers. Uploaded documents stay in memory and are deleted when the session ends. API keys are read from environment variables.

Evaluation and Fallback

- **Retrieval:** Hit@k and MRR on the question set (target: correct section in the top 5 for at least 75%).
- **Answers:** faithfulness checks of the cited sections (target: at least 90% valid citations), a manual check of sample answers, and the refusal rate on out-of-scope questions (target: at least 80%).
- **Document and case features:** accuracy of issue category, dates and amounts, claim labels, contradiction detection, missing-information checks, fact-to-section links, evidence-matrix cells and case summaries on the labelled samples, as set in section 6.1. Weak results will be reported openly.
- **When the AI is wrong or unavailable:** low retrieval confidence produces a 'cannot find a supported answer' message; a failed citation check blocks the output; if the LLM API or the file reader fails, a clear error is shown and no unverified answer appears; unreadable files are reported instead of analysed; if no procedure is stated, the user is referred to a qualified lawyer or legal-aid service.

8. Team Member Responsibility Matrix

Member Name	ID	Primary Responsibilities
Member 1 [name]	[ID]	Data and retrieval: statute collection, cleaning, section splitting, metadata and snapshot documentation; embedding model comparison, vector database, re-ranking and retrieval evaluation; F2 (RAG retrieval), F12 (similar provision / scenario finder) and F15 (case summary generator); scenario library.
Member 2 [name]	[ID]	Generation and analysis: prompt design and LLM integration; F1 (Q&A), F3 (citation verification), F4 (issue extraction), F6 (verification engine), F7 (contradiction

Member Name	ID	Primary Responsibilities
		detector), F11 (evidence status) and F13 (fact-law-evidence mapping); refusal logic and safety checks.
Member 3 [name]	[ID]	Interface, documents and evaluation: frontend and API integration; F5 (document upload and analysis), F8 (missing-information detector), F9 (drafting templates), F10 (next-step guide) and F14 (case evidence matrix); feedback storage; test questions and sample documents; user testing.

All members share documentation, weekly meetings, code review, cross-checking of the statute text used in templates and checklists, system testing, requirement validation and presentation preparation.

Appendix A: Questionnaire

Google Forms questionnaire used to collect user requirements (20 questions, five sections). The last column shows the features the question informs. The questionnaire received 9 responses, all from university or college students (summary in Appendix D).

Q	Question and answer options	Informs
Section A – Background and need		
Q1	Which of the following best describes you? (General citizen / University or college student / Law student / Lawyer / Legal aid volunteer or NGO worker / Other)	Respondent type
Q2	Have you ever needed information about a law or legal rule in Bangladesh? (Yes, multiple times / Yes, once / No, but I may need it / No, never)	Need
Q3	Which sources do you usually use when looking for legal information? (Search engines / Social media / AI chatbots / Government legal websites / Lawyers or experts / Friends or family / Books or legal documents / Other) – select all	Existing practice
Q4	How difficult is it for you to find the correct section of a Bangladeshi law? (Very / Somewhat / Slightly / Not difficult / No experience)	Problem statement

Q	Question and answer options	Informs
Section B – Legal Q&A and statute-grounded answers (F1, F2, F3, F11)		
Q5	If you find the formal language of a law difficult, how important is an explanation in simple Bangla? (1–5)	F1
Q6	How important is it to show the Act name, exact section number and original legal text with an AI answer? (1–5)	F2, F3
Q7	How concerned are you that AI-generated legal answers may be incorrect, outdated or based on another country's laws? (Extremely / Somewhat / Slightly / Not concerned / Not sure)	F3, F11
Section C – Document analysis and verification (F4–F9)		
Q8	How useful would it be to upload a legal document and receive an explanation in simple Bangla? (Extremely … Not useful / Not sure)	F4, F5
Q9	Would it be useful if an assistant checked whether the terms in a document are consistent with relevant laws and gave sources? (Definitely … Not useful at all)	F6
Q10	How useful would it be to detect contradictions such as inconsistent dates, amounts or statements? (Extremely … Not useful / Not sure)	F7
Q11	Would a checklist of missing information or documents relevant to your legal issue be helpful? (Extremely helpful / Somewhat / Not sure / Not helpful)	F8
Q12	Would you use an AI assistant to prepare a preliminary Bangla draft of a complaint or application? (Definitely … Never)	F9
Section D – Legal guidance and advanced features (F10–F15)		
Q13	How important is information about possible next steps or where to seek help, supported by legal provisions? (1–5)	F10
Q14	How useful would it be to receive related laws or sections, relevant examples and a structured case summary in one place? (Extremely … Not useful / Not sure)	F12, F15
Q15	Which features would be most important in a Bangla Legal Assistant? (Simple Bangla explanations / Exact citations and section numbers / Access to original text / Answers with evidence and source indicators / Document analysis and checklists / Bangla drafting / Law-supported next steps / Privacy and document security / Free access / Other) – select all	Feature priority

Q	Question and answer options	Informs
Q16	How useful would it be if the assistant indicated whether an answer is directly supported, partially supported, or needs further verification? (Extremely … Not useful / Not sure)	F11
Q17	Would it help if the assistant organised a legal issue into facts, relevant law and available supporting evidence? (Very helpful … Not helpful at all)	F13
Q18	How useful would a structured evidence matrix be for organising documents, statements, facts and evidence for each fact? (Extremely … Not useful / Not sure)	F14
Section E – Source reliability and user trust		
Q19	How important is it to verify an AI-generated legal answer against the original text of the relevant Act or provision? (1–5)	F3, F11
Q20	What is your biggest concern when using an AI-powered legal information assistant? (Incorrect or misleading information / Missing or inaccurate citations / Outdated provisions / Privacy of uploaded documents / Difficulty understanding the answer / Overreliance on AI instead of professionals / Limited access or cost / No major concerns / Other)	F3, NFR-4

Appendix B: Expert Consultation Record

Record of the consultation with a law faculty member, used as evidence for Section 4.3.

Participant	Samir Bin Shibbir, Lecturer, Daffodil International University (law faculty)
Date and mode	10/10/2026; in person
Material discussed	The proposed features F1–F15 and the scope (statute-grounded answers, document analysis, drafting, next steps, evidence status)
Points raised – good practice	• Show the exact section of the Act that supports the result • Refuse when the law does not support the question • Help the user prepare a draft complaint or application
Points raised – to avoid	• Giving legal advice • Invented (hallucinated) legal content
Other requirements stated	• Results must match the exact section (dhara) of the Act • Information should come with its source • Users should be able to find updates to the law
Outcome	The proposed features were liked; no feature was added, removed or changed. FR-4 and FR-30 were revised (Section 4.4).

Record	Summary of the consultation recorded above; any supporting message or chat screenshot may be added here.

Appendix C: Elicitation Evidence Checklist

Evidence of elicitation included in or referenced by this report.

Done	Item	Where in this report
Yes	Google Forms questionnaire (20 questions)	Appendix A
Yes	Questionnaire response summary (9 responses)	Appendix D; Section 4.3 B
Yes	Record of the consultation with the law faculty member (10/10/2026)	Appendix B; Section 4.3 A
Yes	Requirement change log and confirmed requirements	Section 4.4

Appendix D: Questionnaire Response Summary

Counts of the 9 responses received (all respondents: university or college students). For the 1–5 scales, 5 is the highest.

Q	Topic	Responses (count)
Q1	Respondent type	University / college student 9
Q2	Needed legal information	Yes, multiple times 5; Yes, once 2; No, but may need in future 2
Q3	Sources used	Search engines 5; AI chatbots 3; no answer 1
Q4	Difficulty finding the correct section	Very difficult 3; Somewhat 3; Slightly 2; No experience 1
Q5	Importance of simple Bangla (1–5)	5: 5; 4: 1; 3: 1; 2: 2 (mean 4.0)
Q6	Importance of Act, section and original text (1–5)	5: 5; 4: 2; 2: 2 (mean 4.1)
Q7	Concern about incorrect, outdated or foreign-law answers	Extremely 5; Somewhat 1; Slightly 3
Q8	Usefulness of document upload and Bangla explanation	Extremely 6; Somewhat 1; Slightly 2
Q9	Usefulness of consistency check with sources	Definitely 7; Probably 1; Not sure 1

Q	Topic	Responses (count)
Q10	Usefulness of contradiction detection	Extremely 7; Somewhat 1; Slightly 1
Q11	Helpfulness of missing-information checklist	Extremely 6; Somewhat 3
Q12	Would use Bangla draft assistant	Definitely 5; Probably 4
Q13	Importance of next steps with legal support (1–5)	5: 3; 4: 4; 3: 1; 2: 1 (mean 4.0)
Q14	Usefulness of related laws, examples and case summary	Extremely 8; Slightly 1
Q15	Most important feature (one choice each)	Reliable answers with evidence and source indicators 3; Document analysis and checklists 2; Simple Bangla explanations 2; Exact citations and section numbers 1; All 1
Q16	Usefulness of evidence status indicator	Extremely 7; Somewhat 2
Q17	Helpfulness of facts–law–evidence structure	Very helpful 8; Somewhat 1
Q18	Usefulness of evidence matrix	Extremely 7; Somewhat 1; Slightly 1
Q19	Importance of verifying against original text (1–5)	5: 4; 4: 4; 3: 1 (mean 4.3)
Q20	Biggest concern	Incorrect or misleading information 6; Over-reliance on AI 2; Privacy of uploaded documents 1